

CONSERVATION PRINCIPLES

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1. INTRODUCTION

2. CONSERVATION OF ANYTHING

- Let $\square \subset \mathbb{R}^3$ be an arbitrary volume of space,
- Let $\partial\square = S \subset \mathbb{R}^2$ be the surface enclosing the volume,
- Let $d\mathbf{S} \in \mathbb{R}^3$ be the outward normal vector of a small patch on surface $\partial\square$, and
- Let $\mathbb{T} = [0, T] \subset \mathbb{R}^+$ be the time interval of interest.

Let $f(\mathbf{x}, t) : \square \times \mathbb{T} \mapsto \mathbb{R}^+$ be a function that describes a material property, defined as item per unit of volume $\square$, varies according to current position $\mathbf{x}$ and time t .

There are two special cases:

- $f = f(t)$ is referred to as **uniform** since f does not vary with position $\mathbf{x}$.
- $f = f(\mathbf{x})$ is referred to as **constant** since f does not vary with time t .

Two equivalent viewpoints are explicated: Eulerian and Lagrangian. With Eulerian, the volume $\square$ is fixed in space for all time. With Lagrangian, the volume $\square$ moves through space and time.

2.0.1. *Eulerian formulation.* Let point $\mathbf{p}(\mathbf{x}) : \square \mapsto \mathbb{R}^3$ be a point fixed in space for all time. Point $\mathbf{p}$ is located by Cartesian coordinates (x, y, z) , thus $\mathbf{p}(\mathbf{x}) = \mathbf{p}(x, y, z)$. Note that $\mathbf{p}$ is not a function of time t . Let the vicinity of volume $\square$ be located by point $\mathbf{p}$.

How can property f be varied within the volume $\square$? It can be varied either from *within* the volume or *across* the volume surface, referred to **source** and **flux**, respectively.

- **Source** (or more precisely, point source) says that at point $\mathbf{x}$, the property f is created (added) or deleted (subtracted) from a point source or sink, respectively. Let source $\dot{f}^*(\mathbf{x}, t)$ denote the property source/sink rate at point $\mathbf{x}$ and time t with dimensions of property per unit volume per unit time.
- **Flux** describes the property moving across the boundary $\partial\square$. Movement can arise in two ways:
 - Absent of fluid transport: Let $\mathbf{q}$ be the flux of f in the absence of fluid transport; for example, with heat conduction.
 - Presence of fluid transport: Let $f\mathbf{v}$ be the transport flux of the property because of fluid transport, with units of property per unit area per unit time.

Finally, we state that property f is *conserved* because changes in time of f across the volume $\square$ are caused by either sources or fluxes. Stated mathematically,

$$\frac{d}{dt} \int_{\square} f \, d\square = \underbrace{\int_{\square} \dot{f}^* \, d\square}_{\text{source}} + \underbrace{\left\{ - \int_{\partial\square} \mathbf{q} \cdot d\mathbf{S} - \int_{\partial\square} f\mathbf{v} \cdot d\mathbf{S} \right\}}_{\text{flux}} \quad (1)$$

Note the minus signs for the flux terms are used with the understanding that movement of the property from inside to outside the volume (positive outward flux) *reduces* the time rate of change of the property.

Next, recall the divergence theorem, also known as Gauss's theorem, relates flow (flux) of a vector field through a surface to the behavior of the vector field inside of the surface. That is, for a volume $V \subset \mathbb{R}^3$ that is compact and has a piecewise smooth boundary S and outward unit normal $\mathbf{n}$, if $\mathbf{F}$ is a continuously differentiable vector field defined in the vicinity of V , then

$$\int_S (\mathbf{F} \cdot \mathbf{n}) dS = \int_V (\nabla \cdot \mathbf{F}) dV \quad (2)$$

Use of the divergence theorem used for the flux terms in (1) gives

$$\frac{d}{dt} \int_{\square} f d\square = \underbrace{\int_{\square} \dot{f}^* d\square}_{\text{source}} - \underbrace{\int_{\square} \nabla \cdot (\mathbf{q} + f\mathbf{v}) d\square}_{\text{flux}} \quad (3)$$

Because the volume $\square$ and surface $\partial\square$ are fixed in an inertial reference frame, the time derivative of summed properties equals the sum of local time derivatives of properties, or

$$\frac{d}{dt} \int_{\square} f d\square = \int_{\square} \frac{\partial f}{\partial t} d\square \quad (4)$$

Use of the foregoing equation in (3) gives

$$\int_{\square} \left(\frac{\partial f}{\partial t} + \nabla \cdot (\mathbf{q} + f\mathbf{v}) \right) d\square = \underbrace{\int_{\square} \dot{f}^* d\square}_{\text{source}} \quad (5)$$

Since the volume $\square$ has arbitrary shape and size, the following must also be true, which is the general form for all conservation laws in continuum mechanics,

$$\frac{\partial f}{\partial t} + \nabla \cdot (\mathbf{q} + f\mathbf{v}) = \dot{f}^* \quad (6)$$

3. THE MATERIAL DERIVATIVE

Let $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ be the function of a fluid particle at time t that was located at position $\mathbf{X}$ at time $t = 0$. Let some scalar property $\mathcal{P}$ (for example, temperature or density) of the fluid be attached the fluid at $\mathbf{X}$ and as this point located by $\mathbf{X}$ moves, the property $\mathcal{P}$ is invariant in time. This means

$$\left. \frac{\partial \mathcal{P}}{\partial t} \right|_{\mathbf{X}} = 0. \quad (7)$$

When a property $\mathcal{P}$ is invariant in time for a particle $\mathbf{X}$ of the fluid, the invariant scalar is *material*.

The Eulerian point of view of this same property $\mathcal{P}$ proceeds by selecting a point $\mathbf{x}$ that is fixed in space, and expressing the material invariance in terms of properties of the fluid *at* point $\mathbf{x}$. In general, the fluid will be moving at this point $\mathbf{x}$, so the velocity $\mathbf{v}$ of the fluid at this point enters through the chain rule. That is

$$\left. \frac{\partial \mathcal{P}(\boldsymbol{\varphi}(\mathbf{X}, t), t)}{\partial t} \right|_{\mathbf{X}} \stackrel{(8)}{=} 0 = \left. \frac{\partial \mathcal{P}}{\partial \varphi_i} \right|_t \left. \frac{\partial \varphi_i}{\partial t} \right|_{\mathbf{X}} + \left. \frac{\partial \mathcal{P}}{\partial t} \right|_{\boldsymbol{\varphi}} = \nabla \mathcal{P} \cdot \mathbf{v} + \mathcal{P}_t. \quad (8)$$

This gives rise to the notation of **material derivative**, written with capital D as

$$\frac{D(\cdot)}{Dt} \triangleq \frac{\partial(\cdot)}{\partial t} + \mathbf{v} \cdot \nabla(\cdot). \quad (9)$$

4. CONSERVATION OF MASS

Conservation of mass is obtained from (7) with the following substitutions:

- $f = \rho(\mathbf{x}, t)$ is the mass per unit volume (or density),
- $\mathbf{q} = \mathbf{0}$ because flux of mass can only occur with transport, and
- $\dot{f}^* = 0$ since mass can be neither created or destroyed.

This gives conservation of mass, also called the **continuity equation**

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (10)$$